

Year 7 · Chapter 5B · Equations: answers (teacher copy)

5.10 Guess, check and improve (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: yes
- Ex 1b: $m = 4$
- Ex 2a: $k = 0$
- Ex 2b: $d = 18$
- Ex 3a: $a = 9$
- Ex 3b: $x = 0.5$

Easy · Set A (quick drill)

- yes
- no
- yes
- yes
- no
- yes
- 8
- 8
- 14
- 4
- 9
- 15

Easy · Set B

- $= 7$
- $= 9$
- $= 17$
- $= 4$
- $= 4$
- $= 20$

Medium · Set C

- $m = 4$
- $k = 0$
- $d = 18$
- $a = 9$
- $b = 7$
- $x = 7$

Challenging · Set D

- $x = 0.5$
- $n = 12$ (or -12)
- $x = 5$ (or -6)
- 8

Extension

- $x \approx 3.7$
- $x \approx 6.6$

Exit ticket

- Level 1: yes
- Level 2: $x = 7$
- Level 3: $y = 20$

5.11 Solving one-step equations (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $w = 35$
- Ex 1b: $e = 12$
- Ex 2a: $p = 3$
- Ex 2b: $x = 12$
- Ex 3a: $y = -7$
- Ex 3b: $k = -2.5$

Easy · Set A (quick drill)

- 35
- 12
- 11
- 20
- 15
- 20
- 5
- 15
- 0
- 20
- 7
- 5
- 21
- 2

Easy · Set B

- 3
- 12
- 9
- 45
- 5
- 30

Medium · Set C

- $y = -7$
- $k = -2.5$
- $m = -15$
- $x = -4$
- $b = 3.5$
- $q = 14$

Challenging · Set D

- $n + 19 = 45$, $n = 26$
- $7n = 84$, $n = 12$
- $n/4 = 11$, $n = 44$
- $n - 23 = 58$, $n = 81$

Extension

- $n = 4$
- $a = 15$

Exit ticket

- Level 1: $n = 22$
- Level 2: $x = 7$
- Level 3: $t = -24$

5.12 Solving two-step equations (booklet p. 11)

11)

WE DO examples (teacher)

- Ex 1a: $t = 2$
- Ex 1b: $y = 3$
- Ex 2a: $m = 68$
- Ex 2b: $x = 9$
- Ex 3a: $m = 10$
- Ex 3b: $x = -4$

Easy · Set A (quick drill)

1. 3
2. 2
3. 4
4. 3
5. 5
6. 5
7. 4
8. 5
9. 4
10. 3
11. 8
12. 18

Easy · Set B

1. 68
2. 9
3. 12
4. 20
5. 12
6. 10

Medium · Set C

1. $m = 10$
2. $k = 7$
3. $x = -4$
4. $x = -3$
5. $d = -3$
6. $x = 6$

Challenging · Set D

1. $x = 12$
2. $y = 9.5$
3. $t = 4$
4. $n = 7$

Extension

1. $a = 4$
2. $x = 4$

Exit ticket

1. Level 1: $x = 6$
2. Level 2: $n = 18$
3. Level 3: $k = 9$

5.13 Equations and problem solving

(booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: $C: 17N = 1003$
2. Ex 1b: $N = 59$
3. Ex 2a: $5N = 180$, $N = 36$ tickets
4. Ex 2b: $5N - 180 = 2100$, $N = 456$ tickets
5. Ex 3a: $4s = 52$, $s = 13$ cm
6. Ex 3b: $n + n + 1 + n + 2 = 72$: 23, 24, 25

Easy · Set A (quick drill)

1. $n + 8 = 20$
2. $5n = 45$
3. $n - 7 = 3$
4. $n/2 = 16$
5. $2n + 1 = 15$
6. $3n - 4 = 11$
7. $n + 11 = 30$, $n = 19$
8. $6n = 42$, $n = 7$
9. $n - 8 = 12$, $n = 20$
10. $n/5 = 9$, $n = 45$

Easy · Set B

1. 12
2. 9
3. 10
4. 32
5. 7
6. 5

Medium · Set C

1. \$2.50
2. 13 cm
3. \$138
4. 12
5. 13 km
6. 5 tickets

Challenging · Set D

1. 23, 24, 25
2. length 16 m, width 8 m
3. 456 tickets
4. \$80

Extension

1. 7.5 weeks
2. 32 and 18

Exit ticket

1. Level 1: $n + 14 = 31$, $n = 17$
2. Level 2: $4b = 62$, $b = \$15.50$
3. Level 3: $2n + 9 = 41$, $n = 16$